

Statement of Purpose

I am currently a Lecturer in the Department of Computer Science and Engineering at the University of Asia Pacific (UAP), where I also completed my B.Sc. and am in the final stage of my M.Sc. in Computer Science, expected to be completed in July 2026. These experiences have shaped my understanding of both the theoretical foundations and practical challenges of research in this field and have motivated me to pursue doctoral-level study.

My academic training at UAP provided a solid grounding in machine learning, deep learning, federated learning, and data-driven systems. During my M.Sc., my research has focused specifically on large language models, with work spanning fine-tuning and evaluation of LLMs. This involves problem formulation, experimental design, model development, and scientific writing, all of which I expect to build upon during a PhD.

My research has resulted in publications in venues including IEEE Access, Engineering Applications of Artificial Intelligence, and Computers and Electrical Engineering, as well as book chapters with Elsevier. These contributions have given me exposure to the peer review process and have helped me develop the habit of situating my work within broader research conversations. I am aware, however, that doctoral research demands a level of depth and originality that I am still working toward, and I see a PhD program as the appropriate environment in which to develop that capacity.

My primary research interests lie in advancing machine learning methods for real-world applications, with particular attention to federated learning, trustworthy AI, and secure and resilient intelligent systems. I am drawn to questions about how machine learning models can be made to operate reliably in distributed, dynamic, or adversarial settings. I also have an interest in the intersection of natural language processing and intelligent system design, which has informed several of my recent projects.

In addition to my academic work, I have gained practical experience through a role at DataSoft Systems Bangladesh Limited, where I contributed to the development of scalable web applications using Java, Spring Boot, and Angular. This experience reinforced my understanding of software

engineering principles and gave me perspective on the gap between research prototypes and deployable systems; a distinction I think is worth carrying into doctoral work.

I am seeking a PhD program where I can develop as an independent researcher under the guidance of faculty working on related problems. I am particularly interested in research environments that take both theoretical rigor and applied impact seriously. I recognize that a PhD requires sustained effort and intellectual humility, and I am prepared to invest in that process.

Following the completion of my PhD, I intend to return to Bangladesh and continue my career in academia and research. I hope to contribute to building stronger research capacity in Computer Science and Engineering in my home country, both through teaching and through continued scholarly work.

I note that my IELTS examination is scheduled for August 1, 2026, and I will submit the official score report as soon as it becomes available.